



Individuals Tiebreaker 2024-2025

Student Name:

Team Name:

Rules and Directions

1. **DO NOT FLIP OR DETACH THIS PAGE UNTIL THE TIEBREAKER BEGINS.**
2. **Congratulations** for scoring well on the Individual Round! This is the CMM 2024-2025 **Individuals Tiebreaker**.
3. There are **4 questions with short-answers**, to be completed in **30 minutes** or less.
4. **You may not collaborate with your team** on the Individuals Tiebreaker. In addition, **no other collaboration, computers, calculators, or other outside aid is permitted.**
5. You are permitted to use ruler and compass but **not** a protractor.
6. On the **back side** of this page: write your answers in the corresponding boxes.
7. Answers must be *reasonably simplified* as described in the **CMM Conventions** document. You will not earn credit for a correct but unsimplified answer.
8. The time limit of this test is **30 minutes**, but you may submit answers at **any time** by handing this sheet of paper to the proctor. Students will be ranked based on the number of correct answers and (among those with the same number of correct answers) their submission time.



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Problems

Problem 1. There are four switches in a row, two of which are on and two of which are off. Every second, George randomly flips one of the switches, turning the state of the switch from on to off or from off to on. What is the expected number of seconds it takes until the state of every switch being on at the same time and the state of every switch being off at the same time have both occurred?

Problem 2. How many ordered pairs of integers (a, b) with $1 \leq a, b \leq 1000$ are there such that $1000 \mid a^2 + b^2$?

Problem 3. Points D , E , and F are randomly chosen on $\triangle ABC$, one on each side. What is the probability that the centroid of $\triangle DEF$ is closer to the centroid of $\triangle ABC$ than to any of the sides of $\triangle ABC$? (Note: The centroid is the intersection of the medians in a triangle.)

Problem 4. Complex numbers a, b, c are such that

$$\left| -\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right| = \left| \frac{1}{a} - \frac{1}{b} + \frac{1}{c} \right| = \left| \frac{1}{a} + \frac{1}{b} - \frac{1}{c} \right| = 1.$$

If $|a| = 2025$, then the minimum value of $|b - c|$ is equal to d . What is d^2 ? (Note: $|z|$ denotes the magnitude of a complex number, or its distance from the origin in the complex plane.)

Answer Submissions

Problem 1. 	Problem 2.
Problem 3. 	Problem 4.